

Ying Xu

North China Electric Power University, Beijing, China
fibrecase@ncepu.edu.cn · fibrecase.github.io · github.com/FibreCase

OBJECTIVE

Prospective PhD student in power systems and wind energy, interested in wind-turbine modeling, renewable-energy integration, and power-system simulation; seeking a fully funded position for Fall 2027.

EDUCATION

North China Electric Power University

2023.09 – 2027.06 (expected)

B.Eng. in New Electric Science and Engineering (wind-energy focus)

- Average: 82.9/100. Top courses: Turbine Aerodynamics 95, Wind Resource Measurement & Assessment 95, Numerical Simulation 90, Reliability Engineering 90; also Circuit Analysis, Electrical Machines, Power Electronics, Control Theory

RESEARCH EXPERIENCE

Direct-Drive PMSG Wind-Turbine Multi-Timescale Modeling & Simulation (in progress)

2026.09 – present

National Key Laboratory of Renewable Energy Electric Systems, NCEPU — funded undergraduate research project, sole researcher (advised by Dong Zhao, Senior Engineer)

- Hierarchical multi-timescale model of a direct-drive PMSG wind turbine — slow layer (rotor aerodynamics, drivetrain, pitch control; seconds scale), middle layer (DC-link voltage / power / MPPT; hundreds of ms), fast layer (current control, electromagnetic dynamics; ms scale) — for system-level grid-integration simulation on the lab's in-house power-system platform
- Reduced-order average-value modeling via sensitivity analysis and singular-perturbation order reduction, benchmarked against WECC Type 4 and IEC 61400-27-1 (Type 4A/4B) reference models
- Completed: systematic literature review (12 papers on Type 4/4A/4B standards, model-order reduction, LVRT, and primary-frequency-response control) and the overall modeling scheme; ongoing: model derivation, parameter tables, platform implementation, and wind-speed-step / voltage-sag / frequency-disturbance case studies

CFD Simulation Researcher

2026.07 – 2026.08

Wind Power Research Center, School of New Energy — independent research project under a PhD researcher's guidance

- External-flow CFD simulations of wind-turbine-relevant aerodynamics — cylinder flow, vortex-induced vibration, and static/dynamic stall of NACA0012 and S809 airfoils — in ANSYS Fluent, with selected cases cross-verified in OpenFOAM; responsible for mesh generation, turbulence-model selection, inflation-layer setup, and post-processing of airfoil force coefficients and stall behavior

AWARDS & HONORS

- **National First Prize**, China Undergraduate Electronic Design Contest (2025); nominated for the TI Cup
- National Third Prize, China Undergraduate Electronic Design Contest — TI Cup Analog Electronic System Design (2024)
- University Merit Student; 2nd-Class University Scholarship; Sci-Tech Innovation Scholarship

SELECTED PROJECT

Monocular-Vision Target Measurement Device — National First Prize

2025.07 – 2025.08

China Undergraduate Electronic Design Contest, National Finals (Hardware & Integration)

- Team system: a single 4K industrial camera measures target dimensions and distance (rectangle/circle/triangle), with end-to-end error < 2 mm at 1.6 m
- My scope — embedded systems and hardware: custom power-monitoring circuit (commercial modules prohibited) tracking real-time and peak consumption (peak 7 W, idle 5 W vs. 10 W limit); MIPI display-module adaptation via a custom adapter board
- System debugging and integration across the RK3576 embedded-Linux platform (Flutter front end), supporting overall system and UI integration

INDEPENDENT PROJECTS

12 V Power-Distribution Monitor

2026.09

- End-to-end power-monitoring system: custom open-source PCB + ESP32-C3 firmware (C/ESP-IDF) sampling an INA226 with shunt-calibrated current/power readings, adaptive 10 Hz / 0.1 Hz sampling by viewer presence
- Dual-path overcurrent protection (hardware ALERT-pin comparator + independent backend threshold) with per-event logging; dual-slot OTA with automatic rollback
- Python backend (FastAPI, SQLite, WebSocket) + web dashboard

Wind-Resource Assessment Toolkit

2026.06

- Python pipeline for wind-farm siting and wind-resource assessment: multi-stage automated QC on SODAR/LIDAR (40–200 m) and met-mast (2–80 m) wind data (range / correlation / trend / missing-value checks)
- Weibull distribution fitting, power-law wind-shear and air-density estimation, wind-energy-density wind roses, and per-height data-availability statistics; unit-tested

NanoScope — Two-Channel Pocket Oscilloscope

2025.06 – 2025.07

- Custom measurement instrument: OPA4197 analog front end (–5 V to +12 V), STM32 ADC+DMA high-speed acquisition, frequency/phase measurement via external timers (50 Hz–200 kHz); ESP32 real-time display + BLE Flutter app

LEADERSHIP

Head of Technology Department, MOS Innovation Lab, NCEPU

2025.05 – present

- Organized STM32 embedded-development training and mentored new members; designed the lab's technical-assessment scheme and led project reviews/defenses for dozens of members

TECHNICAL SKILLS

Simulation	ANSYS Fluent (2D external flow), OpenFOAM
Programming	Python, C, MATLAB
Embedded	STM32 / ESP32 (ADC, DMA, UART, SPI, I ² C, PWM), Rockchip Linux, MIPI display
Tools	Git, Docker, Linux, PCB design, AutoCAD, SolidWorks
Languages	English — IELTS 7.0